VISVESVARAYA TECHNOLOGICAL UNIVERSITY

JNANA SANGAMA, BELGAVI-590018

A MINI PROJECT REPORT ON

"CREDIT CARD FRAUD DETECTION USING MACHINE LEARNING"

Submitted in partial fulfilment of the requirements for the degree of

BACHELOR OF ENGINEERING

In

COMPUTER SCIENCE & ENGINEERING

Submitted by

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Kumbalagodu, Mysuru Road, Bangaluru-560074
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CERTIFICATE

This is to certify that the Mini Project Report Entitled "CREDIT CARD FRAUD DETECTION USING MACHINE LEARNING" is a bonafide work carried out by **ADARSHA GC (1DB22CS003), DINESH S (1DB22CS040), SANKETH N (1DB21CS104) & SUDARSHAN (1DB23CS415)** in partial fulfilment for the award of the degree of Bachelor of Engineering in **Computer Science & Engineering** of Visvesvaraya Technological University, Belagavi, during academic year 2024-2025. All Corrections and Suggestions indicated in the Internal Assessment have been incorporated in the Mini Project Report and it is deposited in the department library. The Mini Project Report hasbeen approved as it satisfies the academic requirements in respect of the Project Prescribed forthe Bachelor of Engineering Degree in Computer Science & Engineering.

Dr. Asha KH	**Dr. K B Shiva Kumar**
Assistant Professor	**Asso. Professor and Head**

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Name of the Examiners and Signatures with date:

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DECLARATION

We, Adarsha GC, Dinesh S, Sanketh N, Sudarshan bearing USN: 1DB22CS003, 1DB22CS040, 1DB22CS193, 1DB23CS415 are the Students of 5th semester, B.E in Computer Science & Engineering, Don Bosco Institute of Technology, hereby declare that the entire work embodied in this Mini Project Work titled, "CREDIT CARD FRAUD DETECTION USING MACHINE LEARNING" has been carried out byus and submitted in partial fulfilment of the award of the degree in Bachelor of Computer Science & Engineering of the Visvesvaraya Technological University, Belagavi, during the academic year 2024-2025. Further the matter embodied in this report has not been submittedpreviously by anybody for the award of any degree equal to this or any other university.

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Place: Bengaluru

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Date:

Place: Bengaluru

ABSTRACT

Credit card fraud has become a major concern in the financial industry, especially with the exponential growth of digital transactions. Detecting fraudulent activities requires sophisticated techniques capable of handling complex data patterns and addressing inherent challenges like class imbalance. In this study, machine learning models including Logistic Regression, Decision Trees, Random Forest, and Gradient Boosting were employed, with all implementations carried out using the scikit-learn (sklearn) library. A critical focus was placed on preprocessing the dataset to improve model performance. For instance, Logistic Regression was applied to a balanced dataset, ensuring that the model could fairly predict both fraudulent and legitimate transactions. For Decision Trees, under sampling techniques were utilized to address class imbalance and prevent overfitting.

Ensemble methods like Random Forest and Gradient Boosting were explored for their ability to capture intricate patterns in data and enhance predictive accuracy. Each model's performance was evaluated based on key metrics such as accuracy, precision, recall, and F1-score. Among these, Gradient Boosting demonstrated exceptional performance, effectively identifying fraudulent transactions due to its iterative approach and ability to optimize decision boundaries. This study highlights the importance of combining effective preprocessing techniques with advanced machine learning models to develop robust fraud detection systems. The insights gained not only underscore the potential of machine learning in enhancing financial security but also provide a foundation for future research aimed at mitigating credit card fraud.

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